Automated AI Review Rating Prediction System

Using Machine Learning and Natural Language Processing on Flipkart Product Reviews

Domain

Natural Language Processing (NLP) & Machine Learning

Dataset

Flipkart Product Review Dataset from Kaggle

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1. Introduction

Customer reviews play a critical role in e-commerce platforms by influencing purchasing decisions and providing valuable feedback to sellers and manufacturers. With the rapid growth of online shopping, manually analyzing thousands of customer reviews has become challenging.

This project aims to develop an **Automated AI Review Rating Prediction System** capable of predicting customer ratings directly from textual reviews. By leveraging Natural Language Processing (NLP) techniques and Machine Learning algorithms, the system learns patterns from customer feedback and automatically predicts ratings on a scale of 1 to 5 stars.

The project uses the Flipkart Product Review Dataset and follows a complete machine learning pipeline including data preprocessing, text normalization, visualization, feature engineering, class balancing, model training, and performance evaluation.

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2. Objectives

The primary objectives of this project are:

- To clean and preprocess customer review data.
- To perform text normalization using NLP techniques.
- To analyze review patterns through data visualization.
- To create a balanced dataset for model training.
- To transform textual data into numerical representations using TF-IDF.
- To train multiple machine learning models for rating prediction.
- To compare model performance and identify the best-performing model.
- To develop an automated system capable of predicting customer ratings from review text.

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3. Dataset Description

Dataset Source

Dataset obtained from:[Flipkart Product Review Dataset (Kaggle)]

Dataset Overview

The dataset contains customer reviews and ratings collected from various products available on Flipkart.

Original Dataset Statistics

Attribute	Value
Total Records	189,874
Total Features	5
Dataset Format	CSV
Rating Scale	1–5 Stars

Dataset Features

Column Name	Description
ProductName	Name of the product
Price	Product price
Rate	Customer rating (1–5)
Review	Customer review text
Summary	Product description

Target Variable

Rate

The target variable represents customer ratings:

Rating	Meaning
1	Very Negative
2	Negative

3	Neutral
4	Positive
5	Very Positive

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4. Methodology

The project follows the workflow shown below:

Dataset Collection
↓
Data Preprocessing
↓
Text Normalization
↓
Feature Engineering
↓
Data Visualization
↓
Balanced Dataset Creation
↓
Train-Test Split
↓
TF-IDF Vectorization
↓
Model Training
↓
Performance Evaluation

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5. Data Preprocessing

5.1 Dataset Loading

The dataset was loaded using the Pandas library.

Operations Performed

- Imported CSV dataset.
- Displayed initial records.
- Inspected data types.
- Checked dataset dimensions.

Initial Dataset Shape

Rows	Columns
189,874	5

5.2 Missing Value Handling

Missing values were identified and removed.

Missing Values

Column	Missing Values
ProductName	0
Price	1
Rate	1
Review	4
Summary	14

Action Taken

- Removed records with missing reviews.
- Removed invalid ratings.
- Removed unnecessary null entries.

5.3 Duplicate Removal

Duplicate records were identified and removed.

Results

Description	Count
Exact Duplicate Records Removed	24,861

Conflicting reviews having the same review text but different ratings were retained because they may represent genuine customer opinions across different products.

5.4 Column Selection

The following columns were removed:

Removed Column	Reason
ProductName	Not required for rating prediction
Price	Not directly relevant
Summary	Product description, not customer opinion

Remaining Columns:

Rate
Review

5.5 Text Cleaning

The review text was cleaned using NLP preprocessing techniques.

Cleaning Operations

- Converted text to lowercase
- Removed punctuation
- Removed special characters
- Removed extra spaces
- Removed URLs
- Removed HTML tags
- Removed emojis

5.6 Text Normalization

Further normalization was applied.

Techniques Used

Technique	Applied
Lowercasing	Yes
Stopword Removal	Yes
Lemmatization	Yes
Stemming	No

Example

Original Review	Processed Review
"This Product is Very Good!!!"	"product good"

5.7 Feature Engineering

Additional numerical features were generated.

Feature	Description
word_count	Number of words
review_length	Length of review text
char_count	Number of characters

5.8 Final Dataset Statistics

Description	Value
Final Records	165,000
Features	7

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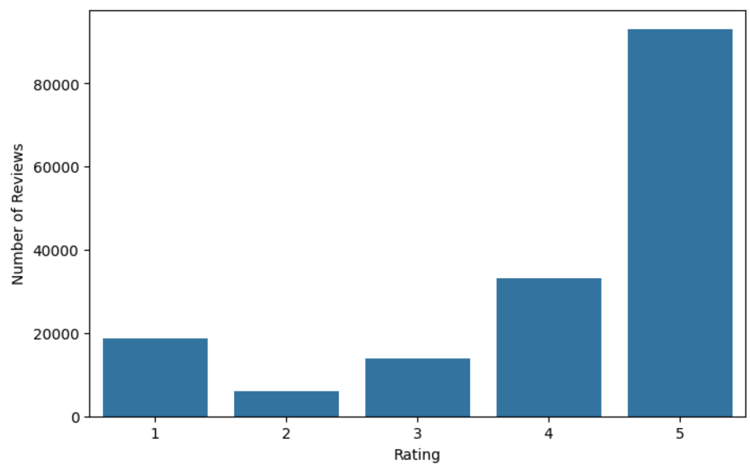
6. Data Visualization

6.1 Rating Distribution

A bar chart was generated to visualize the distribution of ratings.

Figure 1: Rating Distribution Bar Chart



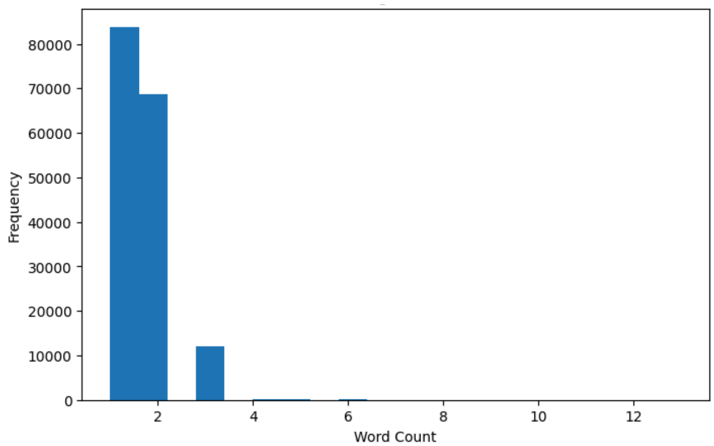
Observation

The dataset is highly imbalanced, with 5-star reviews dominating.

6.2 Review Length Distribution

Histogram of review word counts.

Figure 2: Review Length Histogram



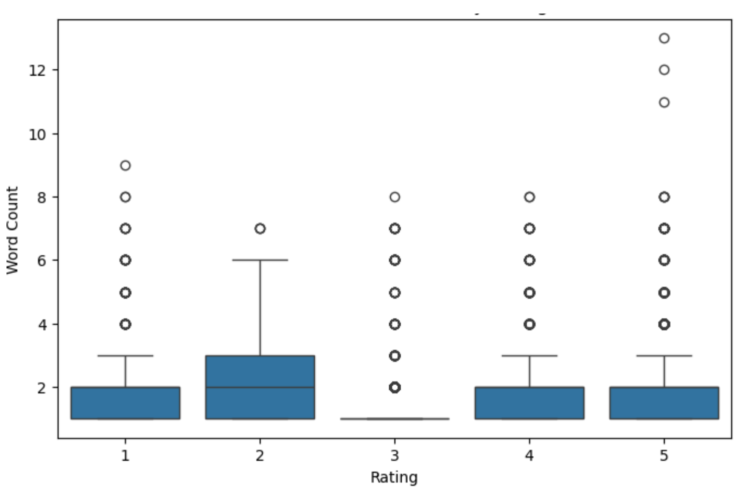
Observation

Most reviews contain only one or two words.

6.3 Box Plot

Comparison of review lengths across rating categories.

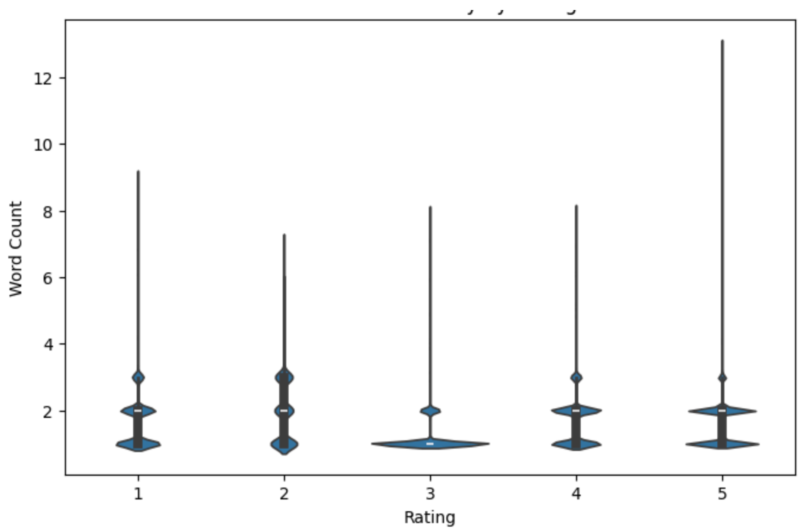
Figure 3: Word Count Boxplot by Rating



6.4 Violin Plot

Distribution density of review lengths.

Figure 4: Violin Plot of Word Count by Rating



6.5 Word Frequency Analysis

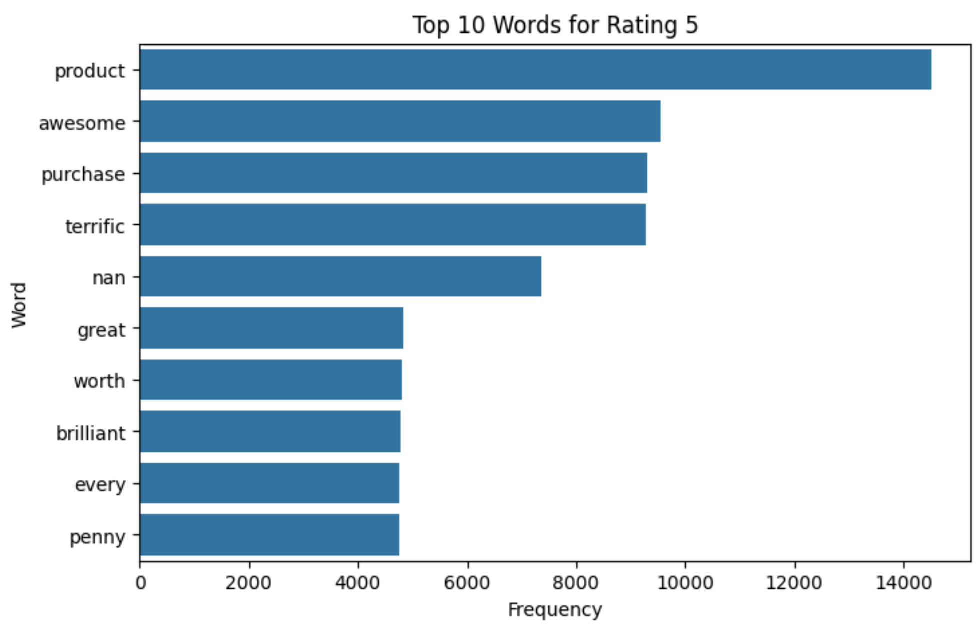
Top frequently occurring words were analyzed for each rating category.

Most Common Words Overall

	Word	Frequency
0	product	25980
1	good	15222
2	nan	15150
3	awesome	9577
4	purchase	9314
5	terrific	9309
6	nice	8158
7	worth	7790
8	wonderful	7763
9	recommended	5732
10	money	5361
11	great	4865
12	brilliant	4795
13	best	4794
14	buy	4788
15	every	4771
16	penny	4770
17	perfect	4770
18	must	4745
19	classy	4736

Table : Top Words for Rating Classes

Rating 1	Rating 2	Rating 3	Rating 4	Rating 5
product : 2320 waste : 2315 money : 2315 absolute : 1231 rubbish : 1230 disappointed : 1197 utterly : 1191 poor : 1189 worthless : 1174	better : 1639 quality : 921 bad : 920 good : 874 product : 868 expected : 856 disappointed : 848 slightly : 845 moderate : 845	product : 2068 nice : 2053 good : 2042 decent : 2025 fair : 2020 okay : 2004 job : 1954 nan : 1721 great : 9 buy : 7	good : 12060 product : 6215 nice : 6020 quality : 3152 choice : 3041 valueformoney : 3032 really : 3022 wonderful : 3007 delightful : 3007 worth : 2982	product : 14509 awesome : 9549 purchase : 9296 terrific : 9290 great : 4840 worth : 4802 brilliant : 4787 every : 4766 penny : 4764



6.6 Sample Reviews

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Rating	Sample Reviews
1	Absolute rubbish! Terrible product Worthless

	Waste of money!
2	Bad quality Value-for-money Moderate Slightly disappointed
3	Just okay Good
4	Nice product Good quality product Worth the money
5	Brilliant Classy product Awesome Must buy!

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7. Balanced Dataset Creation

To address class imbalance, a balanced dataset was created.

Original Distribution

Rating	Count
1	18,782
2	6,014
3	13,945
4	33,207
5	93,052

Balanced Distribution

Each class was resampled to contain 20,000 reviews.

Rating	Count
1	20,000
2	20,000
3	20,000
4	20,000
5	20,000

Total Balanced Records

100,000 Reviews

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8. Train-Test Split and Feature Extraction

Train-Test Split

Stratified splitting was performed.

Dataset	Size
Training Set	80,000
Testing Set	20,000

Split Ratio

80% Training
20% Testing

TF-IDF Vectorization

Text reviews were transformed into numerical feature vectors using TF-IDF.

Advantages

- Captures word importance.
- Reduces influence of common words.
- Suitable for text classification.

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9. Machine Learning Model Development

Three machine learning models were trained:

Model	Type
Multinomial Naive Bayes	Probabilistic
Logistic Regression	Linear Classifier
Random Forest	Ensemble Learning

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10. Model Evaluation and Results

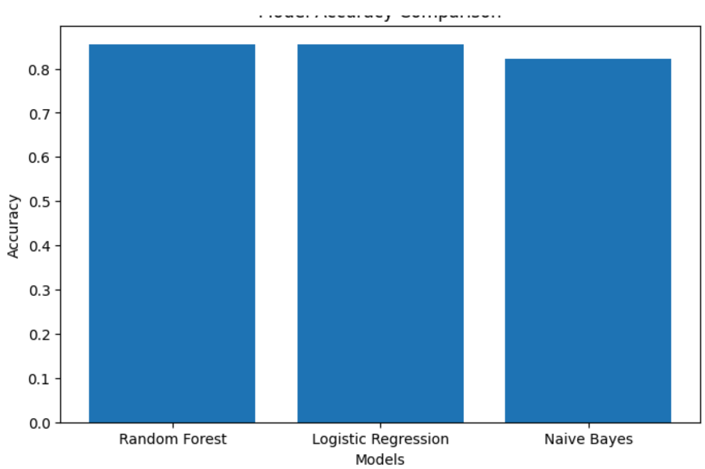
Performance Metrics

- Accuracy
- Precision
- Recall
- F1-Score

Results

Model	Accuracy
Naive Bayes	82.22%
Logistic Regression	85.39%
Random Forest	85.48%

Figure 6: Model Accuracy Comparison



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11. Comparative Analysis

Metric	Naive Bayes	Logistic Regression	Random Forest
Accuracy	82.22%	85.39%	85.48%
Precision	0.86	0.88	0.88
Recall	0.82	0.85	0.85
F1-Score	0.83	0.86	0.86

Best Performing Model

Random Forest achieved the highest accuracy:

85.48%

although Logistic Regression produced nearly identical results.

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12. Conclusion

This project successfully developed an Automated AI Review Rating Prediction System using NLP and Machine Learning techniques. Customer reviews were cleaned, normalized, and transformed into TF-IDF feature vectors. Three machine learning models were trained and evaluated on a balanced dataset.

Among the evaluated models, Random Forest achieved the highest accuracy of 85.48%, demonstrating its effectiveness in predicting customer ratings from textual reviews. The developed system can assist e-commerce platforms in automatically estimating ratings, analyzing customer sentiment, and improving decision-making processes.

Appendix

Final Dataset Files

File	Description
flipkart_reviews_cleaned.csv	Fully cleaned dataset
flipkart_reviews_balanced.csv	Balanced dataset for training

Software and Libraries Used

- Python
- Pandas
- NumPy
- NLTK
- Scikit-Learn
- Matplotlib
- Seaborn
- Google colab